

STREAM: Safe Terrain Recognition & Elevation Analysis for Mars using Deep Learning

Domain: Deep Learning, Computer Vision

1. INTRODUCTION

The exploration of Mars has entered a "New Space" era, with multiple international agencies (NASA, ESA, ISRO, CNSA) planning high-cadence missions ranging from rover deployments to eventual human habitation. A fundamental bottleneck in these missions is the "Entry, Descent, and Landing" (EDL) phase. Selecting a landing site is a high-stakes engineering challenge; hazards such as steep slopes ($>15^\circ$), high surface roughness, and dense boulder fields can result in the total loss of a multi-billion dollar mission.

Traditionally, this selection process is conducted by teams of planetary scientists who manually annotate thousands of high-resolution images. However, as mission frequency increases, manual analysis becomes unsustainable. The **STREAM** (Safe Terrain Recognition & Elevation Analysis for Mars) project proposes an automated, high-precision system leveraging state-of-the-art Deep Learning (DL) to process multimodal satellite data. By fusing 2D optical imagery with 3D elevation models, the system seeks to provide a robust, real-time safety assessment of the Martian surface.

2. PROBLEM STATEMENT

Current methods for landing site selection face three major challenges:

Multimodal Complexity: Difficulty in correlating 2D visual data (images) with 3D physical data (elevation).

Hazard Detection at Scale: Manual identification of craters and boulders over vast regions like the Elysium Planitia is inefficient.

Real-time Feasibility: Lack of automated pipelines that can process raw satellite tiles into actionable hazard maps for mission planning.

3. OBJECTIVES

The primary objectives of this project are:

- To develop a Deep Learning pipeline for semantic segmentation of the Martian surface.
- To fuse multimodal data: **CTX/HiRISE** optical images and **MOLA/HRSC** elevation data.
- To automate "Ground Truth" label generation using physics-based rules (Slope ≤ 7 degree, Low Roughness).
- To evaluate the performance of **UNet/DeepLabv3+** architectures in identifying safe landing spots.
- To produce a visual heatmap overlay for easy interpretation by mission scientists.

4. LITERATURE REVIEW

Recent research in planetary science has shifted from traditional image processing (Hough Transforms for crater detection) to Deep Convolutional Neural Networks (CNNs).

- **CNNs in Space:** Studies have shown that architectures like UNet are highly effective for "pixel-level" classification of terrain.
- **Multimodal Fusion:** Research indicates that using elevation data (DEM) alongside RGB images significantly reduces "false positives" caused by shadows that look like craters.
- **Mars Context:** Existing missions (like InSight and Perseverance) have provided high-resolution datasets that serve as the gold standard for training modern ML models.

5. DATASET & STUDY AREA

Study Area: Landing sites of perseverance and insight for quick verification. This region is chosen due to its volcanic plains and its history as a successful landing site for NASA's Insight mission.

Data Sources:

- **MOLA (Mars Orbiter Laser Altimeter):** Provides the elevation backbone (~460m/pixel).

- **CTX (Context Camera):** Provides medium-resolution imagery (~6m/pixel) for detecting surface texture and craters.
- **Labelling Logic:** Labels are generated automatically where:
 - *Safe (1):* Slope < 7° AND Roughness < Threshold AND No detected craters.
 - *Hazardous (0):* Any pixel failing the above criteria.

6. PROPOSED METHODOLOGY

The project follows a five-stage pipeline:

1. **Data Acquisition:** Fetching GeoTIFF tiles from NASA PDS (Planetary Data System).
2. **Preprocessing:** Aligning CTX images with MOLA DEM. Computation of Slope and Roughness maps using the Rasterio and GDAL libraries.
3. **Data Augmentation:** Creating patches of 256 times 256 pixels with rotation and flipping to improve model generalization.
4. **Model Training:** Implementing a **UNet Architecture** in PyTorch. The model takes a 4-channel input: [Image, Elevation, Slope, Roughness].
5. **Inference & Visualization:** Generating a binary mask of the test area and overlaying it on the original image to show "Green Zones" (Safe) and "Red Zones" (Hazard).

7. TECHNOLOGY STACK

- **Programming Language:** Python 3.12, Jupiter Notebook, Flask
- **Frameworks:** PyTorch / TensorFlow (Keras)
- **Libraries:** OpenCV (Image processing), Rasterio/GDAL (Satellite data), NumPy, Matplotlib.
- **Environment:** Google Colab with Tesla T4 GPU acceleration.VS code

8. EXPECTED OUTCOMES

- A trained Deep Learning model capable of segmenting Mars surface hazards with >85% IoU (Intersection over Union).

- A reusable Python pipeline for downloading and processing NASA Mars data.
- Comparative analysis between different CNN architectures for planetary terrain.
- A final project report and a research-oriented presentation.

9. PROJECT PLAN (GANTT CHART SUMMARY)

- **Month 1:** Literature Review & Environment Setup.
- **Month 2:** Data Acquisition & Preprocessing (Elysium Planitia).
- **Month 3:** Label Generation & Model Development (UNet).
- **Month 4:** Training, Testing, and Performance Tuning.
- **Month 5:** Result Analysis, Heatmap Generation & Final Documentation.

10. REFERENCES

1. NASA Planetary Data System (PDS) - Geosciences Node.
2. Ranneberger et al., "U-Net: Convolutional Networks for Biomedical Image Segmentation."
3. Smith, D. E., et al. "Mars Orbiter Laser Altimeter: Experiment summary after the first year of global mapping of Mars."